

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – X

Roll No.: T043	Name: Hanoch Shetty
Class: TYIT	Batch: 1
Date of Assignment: 8/09/26	Date/Time of Submission: 8/9/26

A. Create a PWA with a web app manifest file including - App name, icons, start URL, display mode, implement push notification in a PWA.

Code:

Program.cs

```
var builder = WebApplication.CreateBuilder(args);
builder.Services.AddRazorPages();
app.MapRazorPages().WithStaticAssets();
app.Run();
self.addEventListener('install', event => {
    console.log('Service Worker Installed');
});
self.addEventListener('activate', event => {
    console.log('Service Worker Activated');
});
```

app.js

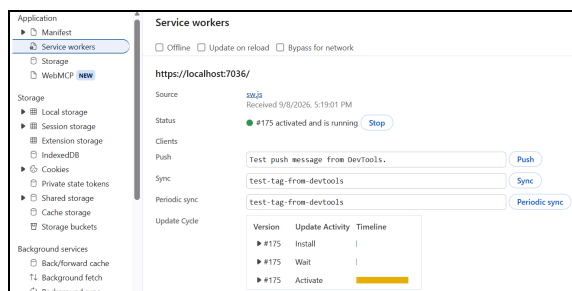
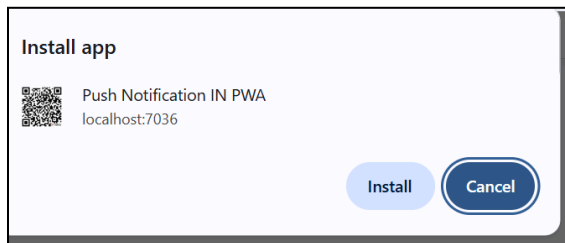
```
if ('serviceWorker' in navigator) {
    navigator.serviceWorker.register('service-worker.js')
        .then(() => console.log('Service Worker
            Registered'))
    function showNotification() {
        Notification.requestPermission().then(() => {
            navigator.serviceWorker.ready.then(Registration {
                registration.showNotification('PWA Notification', {
                    body: 'Push Notification working successfully',
                    icon: 'icons/icon1.png' });
            });
        });
    }
}
```

```
{
    "name": "Push Notification IN PWA",
    "short_name": "Push Notification",
    "start_url": "/index.html",
    "display": "standalone",
    "background_color": "#ffffff",
    "theme_color": "#0d6efd",
    "icons": [
        {
            "src": "icons/icon1.png",
            "sizes": "192x192",
            "type": "image/png"
        },
        {
            "src": "icons/icon2.png",
            "sizes": "192x192",
            "type": "image/png"
        }
    ]
}
```

Layout.cshtml

```
<link rel="manifest" href="~/manifest.json" />
```

Output:



B. Develop an E-commerce PWA with product listing and offline cart functionality (ASP.NET Core Empty Web app)

Code:

Program.cs

```
var builder = WebApplication.CreateBuilder(args);
var apps = builder.Build();

app.UseDefaultFiles();
app.UseStaticFiles();
app.Run();

app.js
if ('ServiceWorker' in navigator) {
    navigator.addServiceWorker('service-worker.js')
        .then(() => { console.log('Service Worker
            Registered') })
let cart = JSON.parse(localStorage.getItem('cart')) || []
displayCart();

function addToCart(product) {
    cart.push(product);
    localStorage.setItem('cart', JSON.stringify(cart), displayCart());
}
function displayCart() {let cartList =
    document.getElementById("cartList");
    cartList.innerHTML = "";
    cart.forEach(item => {
        let li = document.createElement('li');
        li.textContent = item;
        cartList.appendChild(li); }); }
```

service-worker.js

```
const CACHENAME = 'ecommerce-pwa-cache-v1';
const urlsToCache = [
    '/',
    '/index.html',
    '/app.js',
    '/manifest.json'
];

self.addEventListener('install', event => {
    event.waitUntil(
        caches.open(CACHENAME)
            .then(cache => {
                return cache.addAll(urlsToCache);
            })
    );
});

self.addEventListener('fetch', event => {
    event.respondWith(
```

Index.html

```
<!DOCTYPE html>
<html>
<head>
    <meta charset="utf-8" />
    <title>E-Commerce App</title>
    <link rel="manifest" href="manifest.json" />
    <meta name="theme-color" content="#0d63fd" />
    <style>
        body{
            font-family:Arial;
            margin:20px;
        }
        .container{
            display:flex;
            gap:30px;
        }
        .cart-section{
            width:30%;
        }
        .products{
            display:flex;
            flex-wrap:wrap;
            gap:15px;
        }
        .products{
            display:flex;
            flex-wrap:wrap;
            gap:15px;
        }
        .product{
            border:1px solid gray;
            padding:10px;
            width:200px;
        }
        button{
            padding:5px 10px;
            cursor:pointer;
        }
    </style>
</head>
<body>
    <h1>E-Commerce App</h1>
    <div class="container">

        <div class="cart-section">
```

```

        caches.match(event.request)
        .then(response => {
            return response || fetch(event.request);
        })
    );
});

```

Manifest.json

```

{
  "name": "E-Commerce App",
  "short_name": "ShopPWA",
  "start_url": "/index.html",
  "display": "standalone",
  "background_color": "#0d6efd",
  "icons": [
    {
      "src": "../icons/icon1.png",
      "sizes": "128x128",
      "type": "image/png"
    },
    {
      "src": "../icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}

```

```

<h2>Cart Items</h2>
<ul id="cartList"></ul>
</div>
<div class="product-section">
  <h2>Products</h2>
  <div class="product">
    <h3>Laptop</h3>
    <p>Price: ₹50000</p>
    <button onclick="addToCart('Laptop')">
      Add to Cart
    </button>
  </div>
  <div class="product">
    <h3>Smartphone</h3>
    <p>Price: ₹20000</p>
    <button onclick="addToCart('Smartphone')">
      Add to Cart
    </button>
  </div>
  <div class="product">
    <h3>Headphones</h3>
    <p>Price: ₹2000</p>
    <button onclick="addToCart('Headphones')">
      Add to Cart
    </button>
  </div>
</div>
<script src="app.js"></script>
</body>
</html>

```

Output:

E-Commerce App

Cart Items

- Laptop
- Smartphone
- Laptop
- Headphones

Products

Laptop Price: ₹50000 Add to Cart
Smartphone Price: ₹20000 Add to Cart
Headphones Price: ₹2000 Add to Cart